

Two modes: grow a lump sum at a compound rate, or grow a monthly savings plan. Returns the future value, what you actually paid in, and how much of the total is growth.

WHAT IT ASKS YOU

1. Mode: 1 for a lump sum, 2 for monthly deposits
2. Mode 1: Principal, Rate % per year, Times/yr compounded (12 = monthly), Years
3. Mode 2: Monthly deposit, Rate % per year, Years (always compounded monthly)

WHAT IT SHOWS

- FV = future value, 2 decimals
- Grown = the interest part
- Paid = total deposited (mode 2 only)

THE MATH

```
Lump sum:
FV = P (1 + r/100n)^(nt)
Monthly deposits, i = r/1200, m = 12t:
FV = D ((1+i)^m - 1) / i
at r = 0: FV = D * m
```

GET IT ONTO THE CALCULATOR

1. Open **connectevo.ti.com** in Chrome (v143+) and accept the terms.
2. Plug the calculator in with a USB-C *data* cable and switch it on.
3. Click **CONNECT TO CALCULATOR**, pick your calculator, click **Connect**.
4. Click **SEND TO CALCULATOR** and choose COMPINT.py.
5. Check the name reads COMPINT, not PYTHON01. Send.

RUN IT

apps → **Python App** → highlight COMPINT → **Run**. Answer each prompt and press **enter**.

CHECK IT WORKS

```
COMPOUND INTEREST
1 lump sum
2 monthly deposits
Mode: 1
Principal: 1000
Rate % /yr: 5
Times/yr: 12
Years: 10
FV      = 1647.01
Grown   = 647.01
```

GOOD TO KNOW

- Enter the rate as a PERCENT: type 5 for 5%, not 0.05.
- Times/yr is the compounding frequency: 1 = annually, 4 = quarterly, 12 = monthly, 365 = daily.
- A zero rate works in both modes. Mode 2 handles it as a special case because the annuity formula would divide by zero.
- Zero years returns the principal unchanged, which is correct.
- No currency symbol is printed - the euro sign is non-ASCII and renders badly on the calculator. The numbers are unitless, so use them for any currency.
- Negative amounts are rejected. Times/yr must be greater than 0.
- Decimal comma works: 1000,00 is accepted.

SOURCE CODE — COMPINT.PY

```
# COMPINT -- compound interest and monthly-savings growth
# TI-84 Evo / Evo-T / CE Python / 83 Premium CE Python
#
# Numeric output only. No CAS. Mode does not matter --
# arithmetic only.
# Enter the rate as a percentage: 5 means 5%, not 0.05.
# No currency symbol: the euro sign is non-ASCII and
# renders badly.

from math import *

def getnum(msg):
    while True:
        s = input(msg)
        if "," in s and "(" not in s:
            s = s.replace(",", ".")
        try:
            return float(eval(s))
        except:
            print("Not a number")

def getnonneg(msg):
    while True:
        v = getnum(msg)
        if v >= 0:
            return v
        print("Cannot be negative")

print("COMPOUND INTEREST")
print("1 lump sum")
print("2 monthly deposits")

while True:
    m = int(getnum("Mode: "))
    if m in (1, 2):
        break
    print("Pick 1 or 2")

if m == 1:
    p = getnonneg("Principal: ")
    r = getnonneg("Rate % /yr: ")
    while True:
        n = getnum("Times/yr: ")
        if n > 0:
            break
        print("Must be > 0")
    t = getnonneg("Years: ")

    fv = p * (1 + r / (100 * n)) ** (n * t)
    print("FV      = {:.2f}".format(fv))
    print("Grown = {:.2f}".format(fv - p))

else:
    d = getnonneg("Monthly dep: ")
    r = getnonneg("Rate % /yr: ")
    t = getnonneg("Years: ")

    i = r / 1200
    k = 12 * t
    if i == 0:
        # The annuity formula divides by i, so zero rate
        # needs its own case.
        fv = d * k
    else:
        fv = d * (((1 + i) ** k - 1) / i)
    print("FV      = {:.2f}".format(fv))
    print("Paid   = {:.2f}".format(d * k))
    print("Grown = {:.2f}".format(fv - d * k))
```

Mode does not matter. COMPINT only does arithmetic and exponentiation, so the DEGREE / RADIAN setting has no effect on it.